



Saint Columban College
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Serpentine

Serpentine 



Medium General Skills Beginner picoMini 2022 Python

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Description

Find the flag in the Python script!

[Download Python script](#)

Hints 

1 2 3 4

Try running the script and see what happens

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 picoCTF{7h3_r04d_l355_7r4v3l3d_ae0b80bd}

Submit Flag

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Challenge: Serpentine

Category: General Skills

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I. Objective

The objective of this challenge is to **analyze a Python script**, understand how it works, and modify it so that it correctly **reveals the hidden flag**.

II. Background

This challenge involves reading code written in **Python**. Programs are made of:

- **Functions** → blocks of code that do a task
- **Variables** → store data
- **Conditions (if/elif)** → decide what happens

Important Concepts That You Should Know:

1. Functions

A function is like a **machine button**:

```
def print_flag():
```

```
    ...
```

It only works if you **press it (call it)**:

```
print_flag()
```

2. XOR Encryption (Used in this challenge)

The flag is not stored directly. It is **hidden using XOR**.

In the code:

```
flag = str_xor(flag_enc, 'enkidu')
```

This means:

- `flag_enc` = encrypted flag



3. Found this function:

```
def print_flag():  
    flag = str_xor(flag_enc, 'enkidu')  
    print(flag)
```

```
def print_flag():  
    flag = str_xor(flag_enc, 'enkidu')  
    print(flag)
```

This function **decrypts and prints the flag**

4. Then examined the main menu:

```
print('a) Print encouragement')  
print('b) Print flag')  
print('c) Quit')
```

```
while True:  
    print('a) Print encouragement')  
    print('b) Print flag')  
    print('c) Quit\n')  
    choice = input('What would you like to do? (a/b/c) ')
```

5. Checked what happens when choosing **option b**:

```
elif choice == 'b':  
    print('\nOops! I must have misplaced the print_flag  
function!\n\n')
```

```
elif choice == 'b':  
    print('\nOops! I must have misplaced the print_flag function! Check my source code!\n\n')
```



Instead of calling the function, it only prints a message

6. Realized that the function **exists** but it is **never called**
7. Fixed the code by replacing that “print(Oops...)” with the function `print_flag()`

```
elif choice == 'b':  
    print_flag()
```

8. Saved the file and Ran the script



9. Typed b

```
Welcome to the serpentine encourager!  
  
a) Print encouragement  
b) Print flag  
c) Quit  
  
What would you like to do? (a/b/c) b  
picoCTF{7h3_r04d_l355_7r4v3l3d_ae0b80bd}  
a) Print encouragement  
b) Print flag  
c) Quit  
  
What would you like to do? (a/b/c)
```

10. The flag has now decrypted and displayed the flag



V. Result

picoCTF{7h3_r04d_l355_7r4v3l3d_ae0b80bd}

```
picoCTF{7h3_r04d_l355_7r4v3l3d_ae0b80bd}
```

VI. Explanation

Okay, let's try to break it down very simply:

Step 1: The Flag is Hidden

This line stores the encrypted flag:

flag_enc = chr(0x15) + chr(0x07) + ...

This looks confusing on purpose — it's encoded data.

Step 2: There is a Decryption Function

```
def print_flag():  
    flag = str_xor(flag_enc, 'enkidu')  
    print(flag)
```

This function:

- Takes the encrypted flag
- Uses the key "enkidu"
- Decrypts it using XOR
- Prints the real flag

Step 3: The Problem

Inside the menu:

```
elif choice == 'b':  
    print("Oops!...")
```

The program **does not call print_flag()**

So even if the flag exists, it never gets shown sooo



Step 4: The Fix

We changed it to:

```
elif choice == 'b':  
    print_flag()
```

Now:

- The function runs
- The flag is decrypted
- The flag is printed

Simple Analogy

Think of it like this:

- The flag = treasure
- print_flag() = key
- The program = door

Originally:

- You press “b” → door says “I lost the key”

After fixing:

- You press “b” → door uses the key → treasure opens

VII. Conclusion

This challenge teaches important beginner skills:

- Always **read the source code carefully**
- Just because code exists doesn't mean it runs
- Functions must be **explicitly called**
- Simple logic mistakes can hide important behavior

By understanding the structure of the program and making a small change, we successfully revealed the hidden flag.

— The Analyst: Hyposelenia